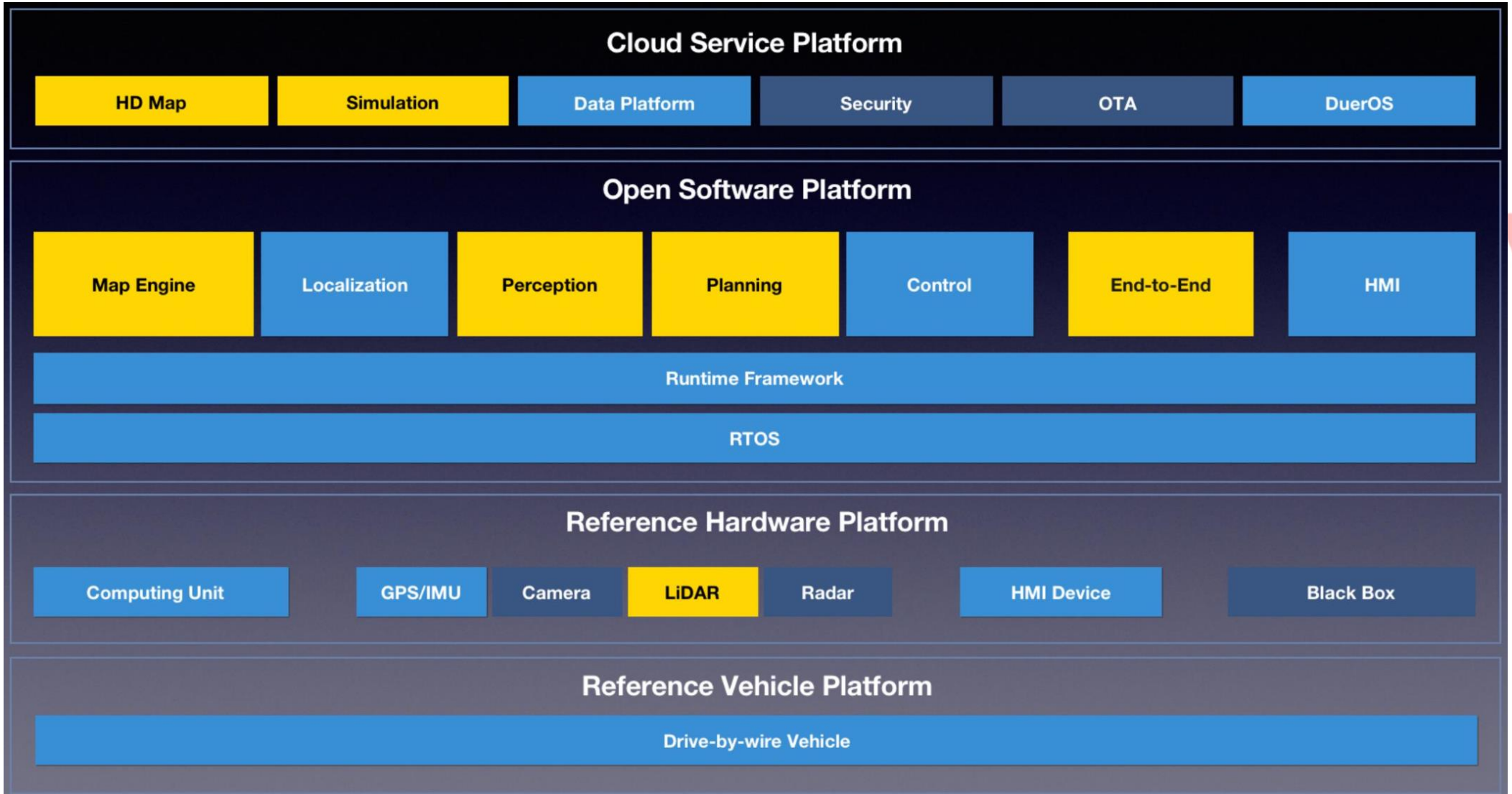
香港城市大學
City University of Hong Kong
Innovating into the Future

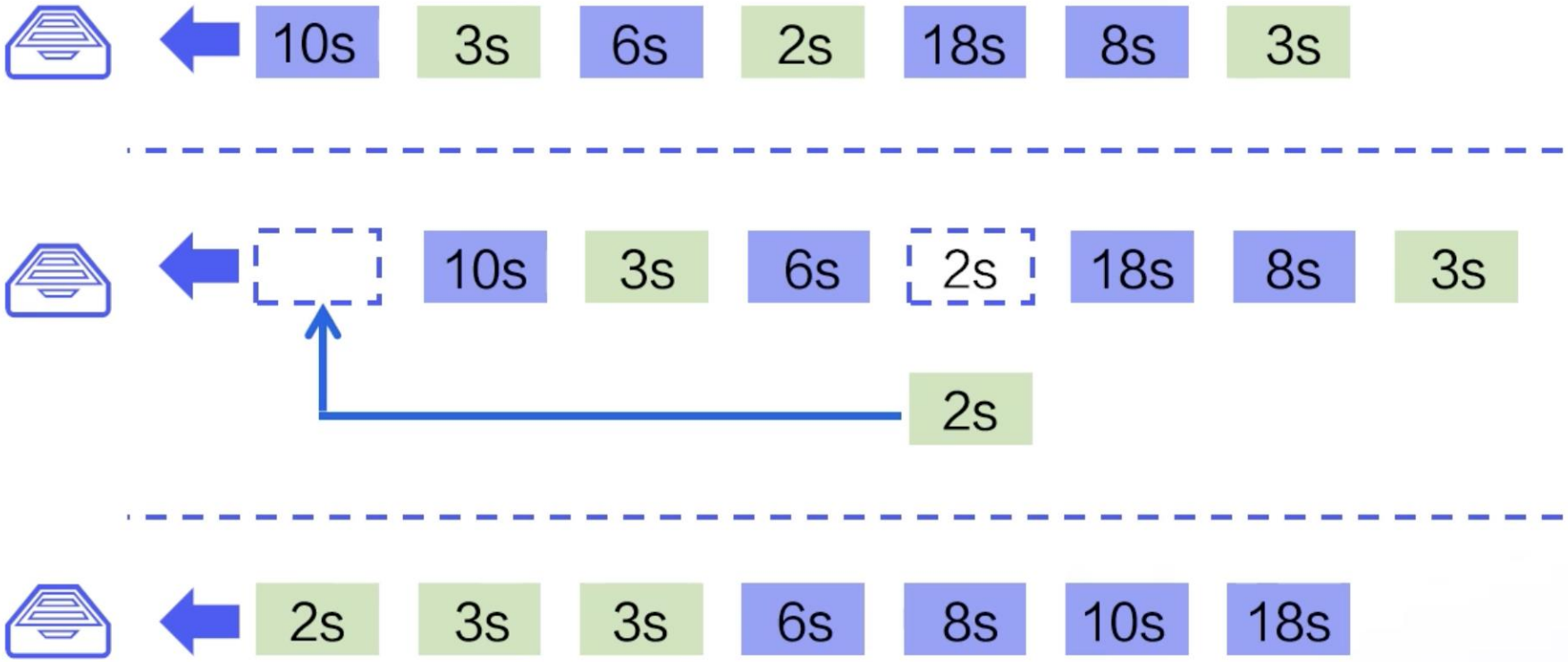
Tutorial 4: Apollo Cyber -- Scheduling and Plugin

Layout of Apollo



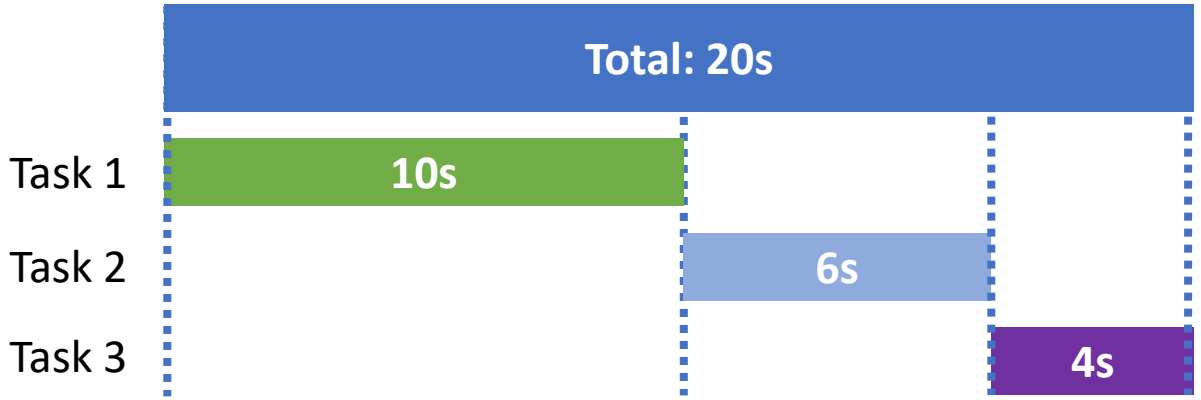
Scheduling in OS

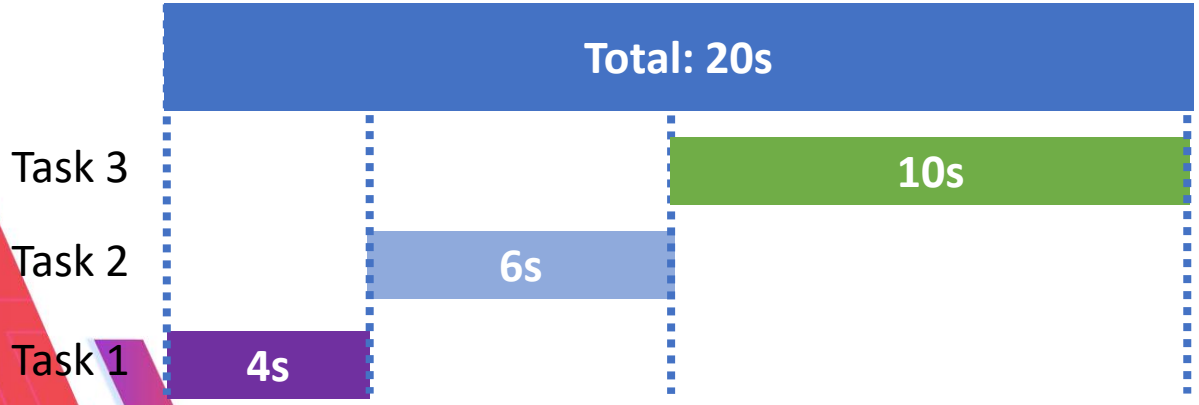
Shortest Job First (SJF) selects the waiting process with the smallest execution time to run first.



Scheduling in OS

Compared with FCFS (first-come-first-served), SJF greatly reduce the total waiting time!



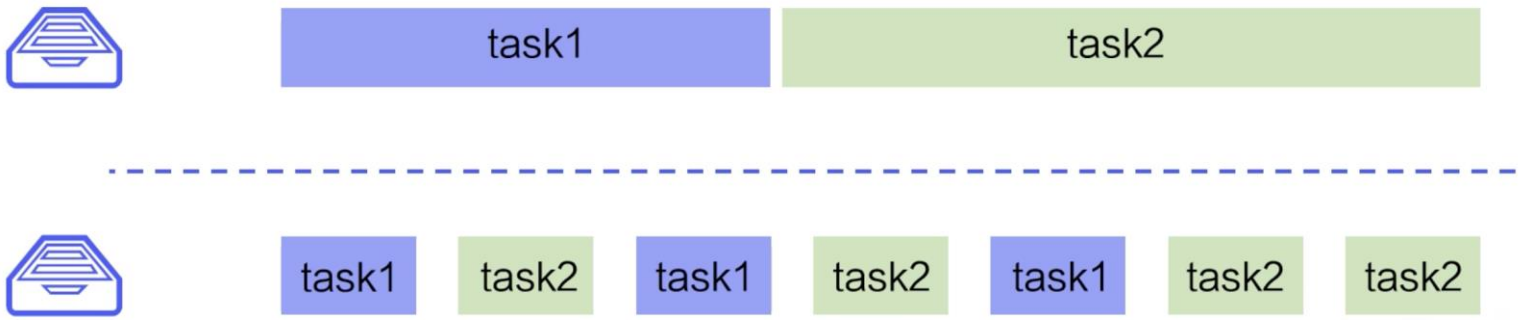


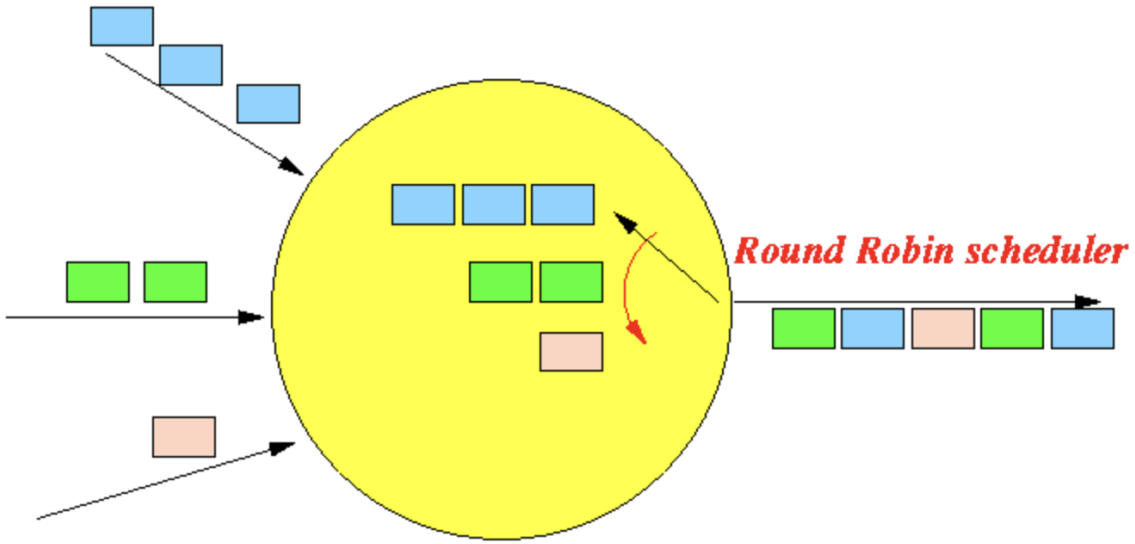
	Total Execution Time(s)	Total Waiting Time(s)
FCFS (First Come First Serve)	20	**26**
SJF (Shortest Job First)	20	**14**

However, in SJF, long time jobs may starve!

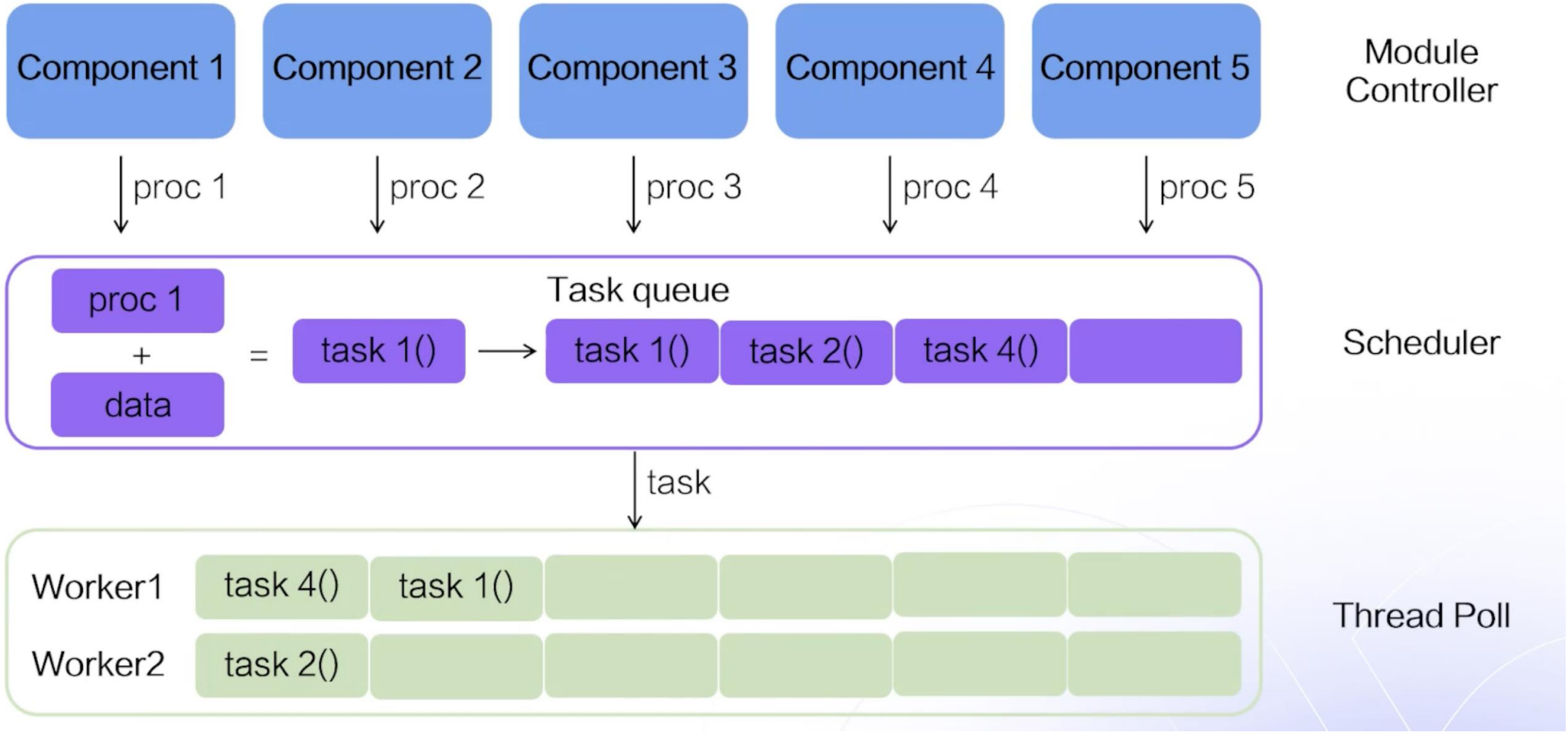
Scheduling in OS

Round Robin (RR) ensure each job can be executed by CPU within a certain period, to avoid the starvation problem





Scheduling in Cyber



Plugin in Cyber

Runtime-Loadable Modules That Extend System Capabilities

Key Characteristics

- **Shared Libraries** loaded at runtime
- **Interface-Based Design** - Implement standard Cyber interfaces
- **Dynamic Discovery** - Automatically recognized by the framework
- **Isolated Functionality** - Self-contained units with clear boundaries

How Plug-ins Work in Cyber

Loading Mechanism:

Key Process:

Framework scans designated plugin directories
Validates plugins against Cyber interfaces
Registers available plugins with corresponding managers
Components can request plugin instances as needed

```
┌──────────────────────────────────────────┐
│      CyberRT Core Framework              │
├──────────────────────────────────────────┤
│        Plugin Manager                    │
│  • Discovers .so files in plugin directory │
│  • Loads at startup or runtime           │
│  • Manages lifecycle                     │
├──────────────────────────────────────────┤
│    Component Factory    │   Data Codec   │
│    └── Plugin A         │   └── Plugin B │
│    └── Plugin C         │   └── Plugin D │
└──────────────────────────────────────────┘
```

Types of Plug-ins in Cyber

Component Plug-ins

- Add new processing algorithms (perception, planning)
- Implement as Component or TimerComponent subclasses

Example: New object detector, custom path planner.

Communication Adapters

- Extend supported message protocols
- Implement Reader/Writer interfaces

Example: ROS bridge, custom serial protocol handler

Data Codec Plug-ins

- Handle custom data serialization formats
- Extend Codec interface

Example: Proprietary sensor format decoder

Tool & Utility Plug-ins

- Debugging, monitoring, visualization tools

Example: Custom data visualizer, performance profiler

Benefits of using Plugin

Runtime Flexibility

- Hot-swapping: Replace algorithms without restarting the system
- Conditional Loading: Load plugins based on hardware availability
- A/B Testing: Easily compare different algorithm implementations

Modular Development

- Independent Teams: Different teams can develop plugins in parallel
- Version Management: Each plugin can have independent versioning
- Selective Deployment: Deploy only needed plugins to each robot

Benefits of using Plugin

Maintenance & Updates

- Zero-Downtime Updates: Update individual plugins without system downtime
- Safe Rollbacks: Quickly revert to previous plugin versions
- Isolated Testing: Test plugins independently of main system

Scalability

- Third-Party Integration: Easily integrate vendor-specific solutions
- Community Contributions: Share and reuse plugins across projects
- Gradual Migration: Phase out legacy systems via plugin wrappers